Tailwind CSS: Zero to Hero Guide

Tailwind is a **utility-first** CSS framework: instead of writing custom CSS rules, you compose small single-purpose classes (`p-4`, `flex`, `text-lg`) directly in your HTML. Every example below shows the code and its live output, rendered by real Tailwind (v3.4) running on this page.

Why | Setup | Spacing | Type & colour | Flex | Grid | Responsive | States | Dark mode | Arbitrary | Motion | Config | Component | Pitfalls | Reference

1. Why Tailwind, and when to choose it

Theory: Tailwind generates only the classes you actually use, so production CSS is tiny (often under 15 KB). Styles live next to the markup, so you never invent class names or hunt through stylesheets, and a design system (spacing, colours, breakpoints) is built in.

Choose Tailwind when…	Choose plain CSS or another approach when…
You build with components (React, Vue, Svelte, templates) and want fast, consistent UI	The project is a tiny static page or you need very unusual, hand-crafted CSS (use plain CSS)
A team needs shared design constraints	You want prebuilt components with a fixed look out of the box (Bootstrap-style libraries)
You want small CSS and no naming debates	Your team strongly prefers semantic class names and separated concerns

2. Setup

Theory: Tailwind scans your files for class names and builds a CSS file with only those. For prototyping, use the CDN script. For production, use the build tool. Version 4 changed the setup: configuration moved into CSS (`@theme`) and there is no required `tailwind.config.js`.

CODE

```
<!-- 1) Quick prototype: CDN (not for production) -->
<script src="https://cdn.tailwindcss.com"></script>

# 2) Production, Tailwind v3 (CLI)
npm install -D tailwindcss@3
npx tailwindcss init                    # creates tailwind.config.js
# tailwind.config.js
module.exports = { content: ["./src/**/*.{html,js,jsx,tsx}"], theme: { e
/* src/input.css */
```

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```
@tailwind base; @tailwind components; @tailwind utilities;
npx tailwindcss -i ./src/input.css -o ./dist/output.css --watch

# 3) Tailwind v4 (Vite)
npm install tailwindcss @tailwindcss/vite     # add tailwindcss() to vit
/* src/input.css */
@import "tailwindcss";
@theme { --color-brand: #4f46e5; }            /* gives bg-brand, text-br
```

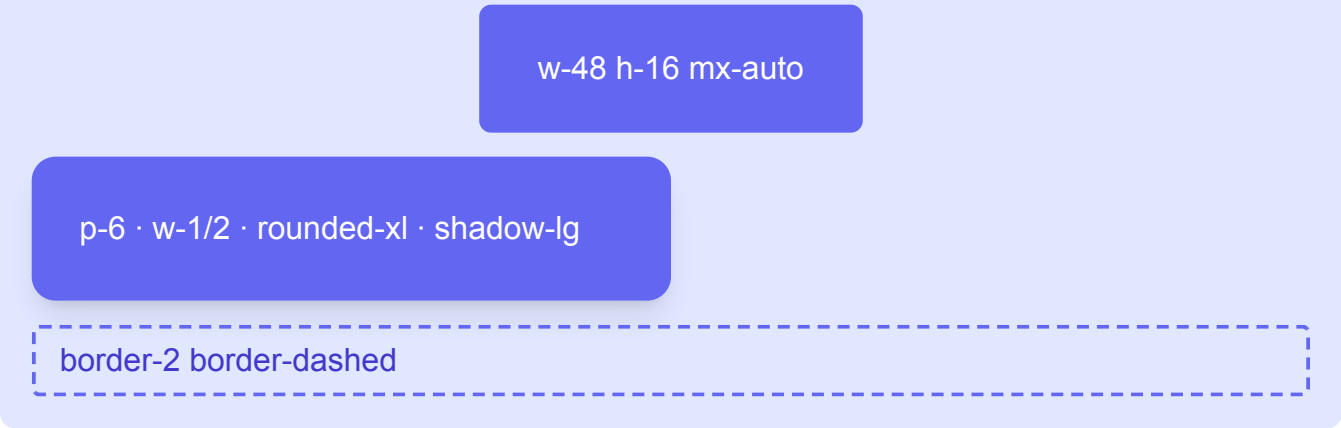
3. Spacing, sizing, borders and shadows

Theory: The spacing scale is multiples of 0.25rem: `4` = 1rem = 16px. `p` padding, `m` margin, `w` width, `h` height; add `x` / `y` / `t` / `b` / `l` / `r` for a side (`px-4`, `mt-2`). Fractions like `w-1/2` and keywords like `w-full` work too. **Choose** `gap` over margins between flex/grid children, and `space-y-*` for stacks.

CODE

```
<div class="bg-indigo-100 p-4 rounded-lg space-y-3">
  <div class="mx-auto flex h-16 w-48 items-center justify-center rounded
  <div class="w-1/2 rounded-xl bg-indigo-500 p-6 text-white shadow-lg">p
  <div class="rounded border-2 border-dashed border-indigo-500 px-3 py-1
</div>
```

OUTPUT



4. Typography and colour

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Theory: Colours use a name plus shade (`slate-600`, 50 to 950). Add an opacity with a slash (`bg-emerald-500/15`). Text utilities: `text-xl` (size), `font-bold`, `leading-relaxed`, `tracking-tight`, `uppercase`, `truncate`. **Choose** `text-slate-600`-type greys for body text and reserve your brand colour for actions.

CODE

```
<h3 class="text-2xl font-extrabold tracking-tight">Heading text</h3>
<p class="mt-1 max-w-prose leading-relaxed text-slate-600 dark:text-slat
<p class="mt-2 text-xs font-semibold uppercase tracking-widest text-indi
<p class="mt-2 text-lg italic underline decoration-pink-500 decoration-w
<span class="mt-2 inline-block rounded-full bg-emerald-500/15 px-2 py-0.
<p class="mt-2 w-40 truncate">This long sentence is cut off with an elli
```

OUTPUT

Heading text

Body text with relaxed leading and a readable maximum width.

SMALL LABEL

Decorated text

emerald-500/15 badge

This long sentence i…

5. Flexbox

Theory: `flex` makes a container; `flex-col` sets direction; `justify-*` aligns on the main axis, `items-*` on the cross axis; `flex-1`, `flex-none`, `shrink-0` control children. **Choose** flex for one-dimensional rows or columns such as navbars, toolbars and button groups.

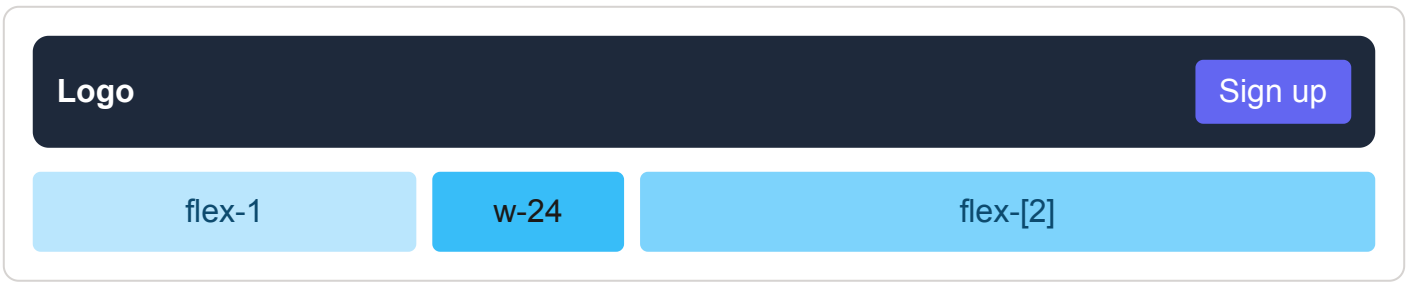
CODE

```
<div class="flex items-center justify-between gap-4 rounded-lg bg-slate-
  <span class="font-bold">Logo</span>
  <nav class="flex gap-4 text-sm"><span>Home</span><span>Docs</span><spa
  <button class="rounded bg-indigo-500 px-3 py-1 hover:bg-indigo-400">Si
</div>
<div class="mt-3 flex flex-col gap-2 text-center sm:flex-row">
  <div class="flex-1 rounded bg-sky-200 p-2 text-sky-900">flex-1</div>
  <div class="w-24 flex-none rounded bg-sky-400 p-2">w-24</div>
```

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```
  <div class="flex-[2] rounded bg-sky-300 p-2 text-sky-900">flex-[2]</di
</div>
```

OUTPUT



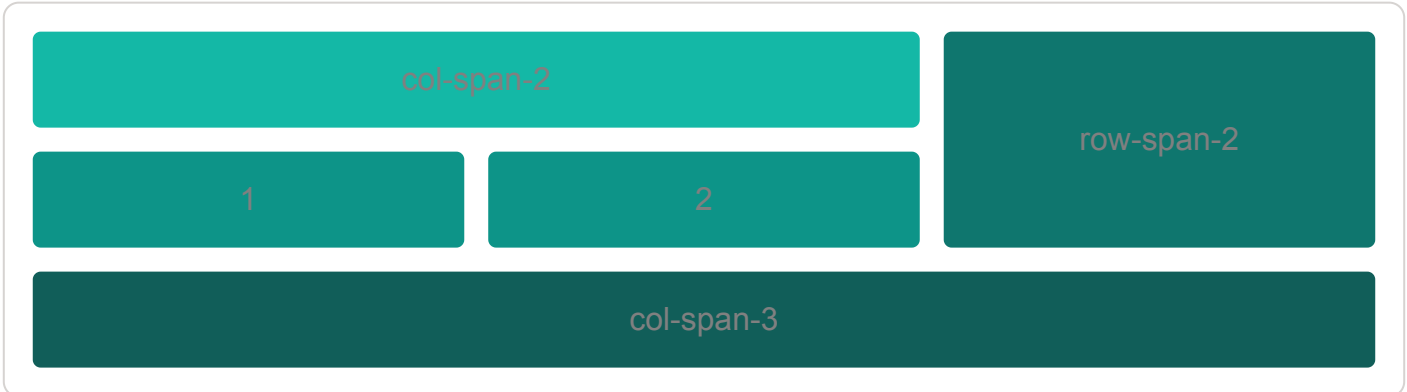
6. Grid

Theory: `grid grid-cols-3` makes three equal columns; `col-span-2` and `row-span-2` stretch items; `auto-rows-*`, `place-items-center` and `gap-*` refine it. **Choose** grid when you need rows and columns together (galleries, dashboards, page layouts).

CODE

```
<div class="grid grid-cols-3 gap-3 text-center text-white">
  <div class="col-span-2 rounded bg-teal-500 p-3">col-span-2</div>
  <div class="row-span-2 grid place-items-center rounded bg-teal-700 p-3
  <div class="rounded bg-teal-600 p-3">1</div>
  <div class="rounded bg-teal-600 p-3">2</div>
  <div class="col-span-3 rounded bg-teal-800 p-3">col-span-3</div>
</div>
```

OUTPUT



7. Responsive design

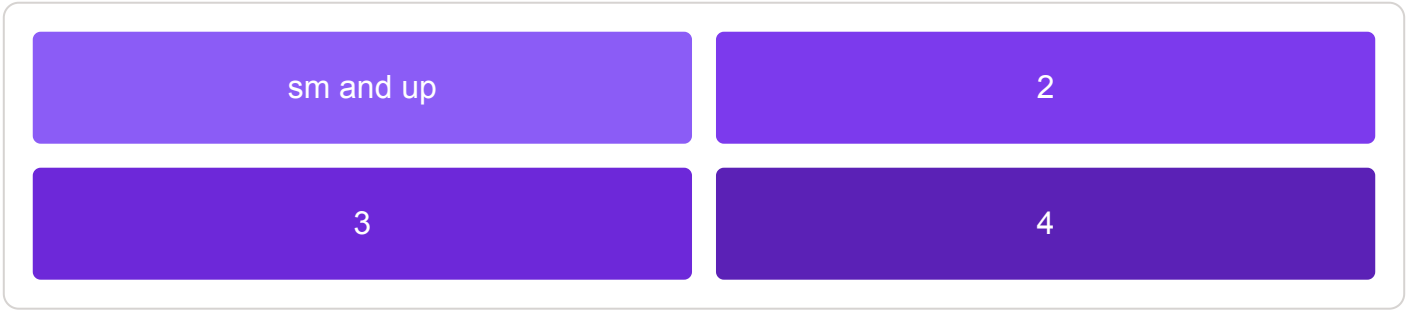
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Theory: Tailwind is **mobile-first**: unprefixed utilities apply to all sizes, and prefixes apply from that width *up*: `sm:` 640px, `md:` 768px, `lg:` 1024px, `xl:` 1280px, `2xl:` 1536px. Use `max-md:` to target below a breakpoint and `@container` plugin utilities for component-level responsiveness. **Choose** to design the small layout first, then add prefixes only where it needs to change. Resize your window to watch this change.

CODE

```
<div class="grid grid-cols-1 gap-3 sm:grid-cols-2 lg:grid-cols-4">
  <div class="rounded bg-violet-500 p-4 text-center text-white"><span cl
  <div class="rounded bg-violet-600 p-4 text-center text-white">2</div>
  <div class="rounded bg-violet-700 p-4 text-center text-white">3</div>
  <div class="rounded bg-violet-800 p-4 text-center text-white">4</div>
</div>
```

OUTPUT



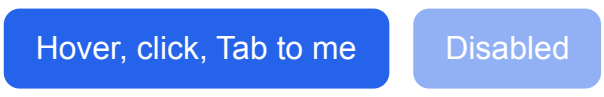
8. State variants: hover, focus, group, peer

Theory: Put a variant prefix before any utility: `hover:`, `focus:`, `focus-visible:`, `active:`, `disabled:`, `first:`, `odd:`. Mark a parent `group` to style children with `group-hover:`; mark an earlier sibling `peer` to style later siblings with `peer-checked:`. Variants stack: `md:hover:bg-red-500`. **Choose** `focus-visible` over `focus` for rings, so mouse users are not distracted.

CODE

```
<button class="rounded-lg bg-blue-600 px-4 py-2 text-white transition ho
<button disabled class="ml-2 rounded-lg bg-blue-600 px-4 py-2 text-white
<div class="group mt-3 rounded border border-slate-300 p-3 hover:border-
<label class="mt-3 flex items-center gap-2"><input type="checkbox" class
<ul class="mt-3 text-sm"><li class="px-2 first:font-bold odd:bg-slate-50
```

OUTPUT



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Hover the card: group-hover

Tick me: peer-checked

first: bold
odd and even rows
odd: tinted

9. Dark mode

Theory: Add `dark:` variants. By default they follow the operating system (`prefers-color-scheme`). To add your own toggle, set `darkMode: 'class'` in the config and put the `dark` class on `<html>`. **Choose** the default media strategy for simple sites and the class strategy when users can switch themes. Change your device theme to see the box below flip.

CODE

```
<div class="rounded-lg border border-slate-200 bg-white p-4 text-slate-8
  <p class="font-semibold">Adaptive card</p>
  <p class="text-sm text-slate-500 dark:text-slate-400">Light by default
</div>
```

OUTPUT

Adaptive card

Light by default, dark when your system is dark.

10. Arbitrary values and variants

Theory: When the scale lacks what you need, use square brackets: `w-[137px]`, `bg-[#1da1f2]`, `grid-cols-[1fr_2fr]` (underscores become spaces), `h-[calc(2rem+8px)]`, or a custom selector like `[&>p]:text-pink-600`. **Choose** a theme token first; use brackets for true one-offs, and add repeated values to the config.

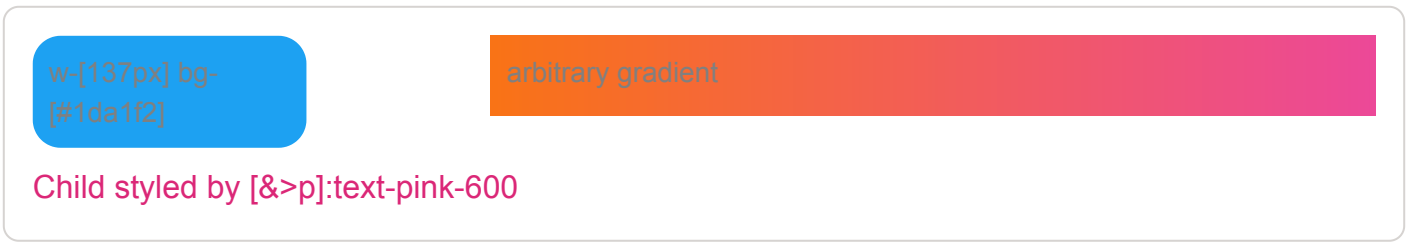
CODE

```
<div class="grid grid-cols-[1fr_2fr] gap-2 text-sm text-white">
  <div class="w-[137px] rounded-[14px] bg-[#1da1f2] p-2">w-[137px] bg-[#
  <div class="h-[calc(2rem+8px)] bg-[linear-gradient(90deg,#f97316,#ec48
```

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```
</div>
<div class="mt-2 [&>p]:text-pink-600"><p>Child styled by [&amp;&gt;p]:te
```

OUTPUT



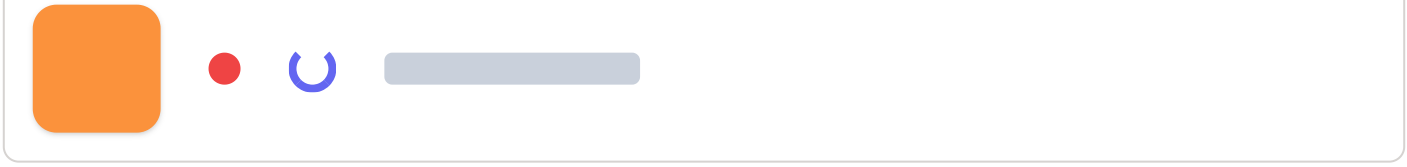
11. Transitions, transforms and animation

Theory: `transition` animates changes; tune with `duration-300` and `ease-out`. Transforms: `scale-110`, `rotate-6`, `-translate-y-2`. Built-in animations: `animate-spin`, `animate-ping`, `animate-pulse`, `animate-bounce`. **Choose** transform and opacity for smooth motion, and wrap motion in `motion-reduce:` variants for users who prefer less.

CODE

```
<div class="flex flex-wrap items-center gap-6">
  <div class="h-16 w-16 rounded-xl bg-orange-400 shadow transition durat
  <div class="h-4 w-4 rounded-full bg-red-500 motion-safe:animate-ping">
  <div class="h-6 w-6 animate-spin rounded-full border-4 border-indigo-5
  <div class="h-4 w-32 animate-pulse rounded bg-slate-400/50"></div>
</div>
```

OUTPUT



12. Customising: config, @apply and components

Theory: Extend the design system in `tailwind.config.js` (v3) or `@theme` (v4). `@apply` bundles utilities into a class. **Choose** a framework component (React, Vue, a template partial) before `@apply`, because components reuse markup and styles together; use `@apply` for third-party HTML you cannot edit or very small repeated patterns.

CODE

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```
/* tailwind.config.js (v3) */
module.exports = {
  content: ["./src/**/*.{html,js}"],
  darkMode: "class",
  theme: {
    extend: {
      colors: { brand: { 500: "#4f46e5", 600: "#4338ca" } },
      fontFamily: { sans: ["Inter", "system-ui", "sans-serif"] },
      screens: { "3xl": "1800px" },
    },
  },
  plugins: [require("@tailwindcss/typography"), require("@tailwindcss/fo
};

/* input.css */
@layer components {
  .btn-primary { @apply rounded-lg bg-brand-500 px-4 py-2 font-semibold
}
```

13. Putting it together: a profile card

Theory: A real component combines everything: layout (flex), spacing, type, colour, states, and dark mode.

CODE

```
<article class="mx-auto max-w-sm rounded-2xl border border-slate-200 bg-
  <div class="flex items-center gap-3">
    <div class="grid h-12 w-12 shrink-0 place-items-center rounded-full
    <div>
      <h3 class="font-semibold text-slate-900 dark:text-white">Asha Roy<
      <p class="text-sm text-slate-500">Product Designer</p>
    </div>
    <span class="ml-auto rounded-full bg-emerald-100 px-2 py-0.5 text-xs
  </div>
  <p class="mt-4 text-sm leading-relaxed text-slate-600 dark:text-slate-
  <div class="mt-4 flex gap-2">
    <button class="flex-1 rounded-lg bg-indigo-600 py-2 text-sm font-sem
    <button class="flex-1 rounded-lg border border-slate-300 py-2 text-s
```

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```
  </div>
</article>
```

OUTPUT

AR

Asha Roy

Product Designer

Online

Designs calm, accessible interfaces for fast-moving teams.

Follow | Message

14. Best practices and pitfalls

- **Never build class names dynamically.** `` `bg-${color}-500` `` is not detected by the scanner. Write full names (`bg-red-500`) or map values to complete strings.
- **Extract components, not classes.** Repeat markup in a component; keep utilities in it.
- **Set `content` paths correctly** (v3), or styles will silently be missing in production.
- **Keep a consistent class order** with the official Prettier plugin (`prettier-plugin-tailwindcss`).
- **Use the scale first,** brackets second; add repeated one-offs to the theme.
- **Mind accessibility:** keep `focus-visible` rings, check colour contrast, and use real elements (`button`, `nav`).
- **Avoid the CDN in production;** it compiles in the browser and is only for prototypes.

15. Quick reference

Group	Common utilities and values
Spacing scale	1 = 4px, 2 = 8px, 3 = 12px, 4 = 16px, 6 = 24px, 8 = 32px, 12 = 48px, 16 = 64px
Breakpoints	sm 640, md 768, lg 1024, xl 1280, 2xl 1536 (min-width, mobile-first)
Font size	text-xs 12, sm 14, base 16, lg 18, xl 20, 2xl 24, 3xl 30, 4xl 36 px
Font weight	font-light 300, normal 400, medium 500, semibold 600, bold 700, extrabold 800
Radius	rounded-sm, rounded (4px), md 6, lg 8, xl 12, 2xl 16, full
Shadow	shadow-sm, shadow, shadow-md, shadow-lg, shadow-xl, shadow-2xl
Display	block, inline-block, hidden, flex, inline-flex, grid, contents
Position	static, relative, absolute, fixed, sticky; inset-0, top-4, z-10

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Sizing	w-full, w-1/2, w-screen, min-h-screen, max-w-md, size-10
Overflow	overflow-hidden, overflow-auto, overflow-x-scroll

Next steps: build a page with the Tailwind docs open (tailwindcss.com/docs), learn `@container` queries, the typography and forms plugins, and component libraries built on Tailwind such as shadcn/ui and Headless UI.

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